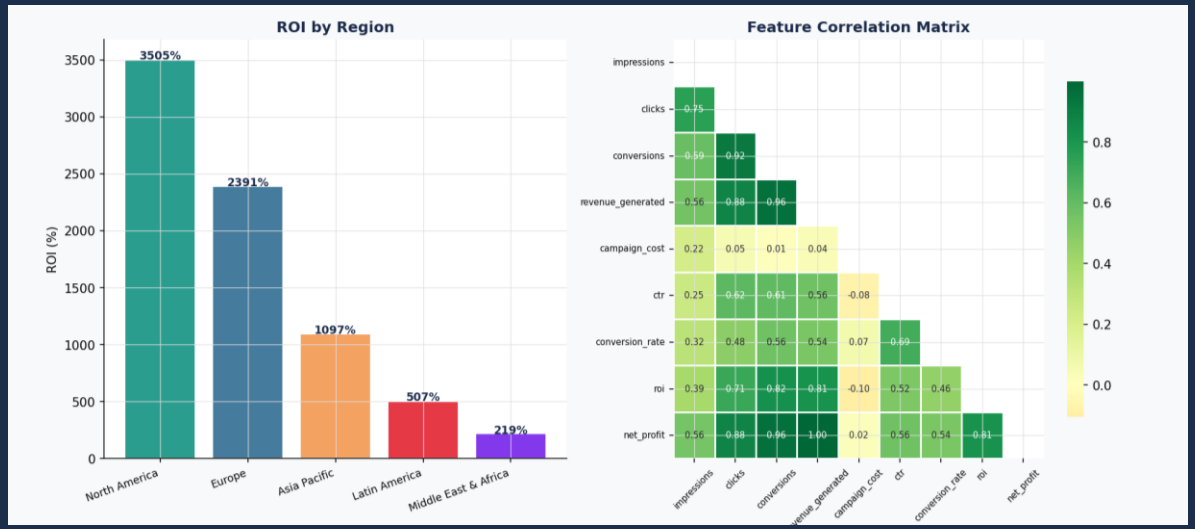
Marketing Campaign ROI



5,000+	6	5	10+
Campaigns	Channels	Regions	DAX Measures

- SQL
- Python
- DAX
- Power Query
- Star Schema

1. Business Problem Definition

Business Objective

Optimize marketing spend across 6 channels and 5 regions by measuring ROI, identifying budget waste, and delivering data-driven reallocation recommendations to the CMO and CFO.

Key Stakeholders

CMO	Channel strategy & budget authority
CFO	Cost efficiency & P&L impact
Marketing Mgr	Campaign-level performance
Regional Lead	Territory revenue targets
Data Team	Pipeline & model governance

7 Critical Business Questions

Q1 What is the overall ROI across all campaigns?

Q2 Which channels & types deliver the highest ROI?

Q3 Where is budget being wasted (negative ROI)?

Q4 How does performance vary by region?

Q5 What are the key drivers of conversion rate?

Q6 How do campaigns perform across time — seasonality?

Q7 Which campaigns should be scaled, optimized, or cut?

2. Data Schema & Star Model Design

Fact table + 4 dimension tables — the foundation for every DAX measure

DIMENSION

dim_date

- **date_id (PK)**
- full_date
- year
- quarter
- month
- month_name
- day_of_week
- is_weekend

FACT TABLE

fact_campaign

- **campaign_id (PK)**
- **date_id (FK)**
- **channel_id (FK)**
- **region_id (FK)**
- **type_id (FK)**
- impressions
- clicks
- conversions
- revenue_generated
- campaign_cost
- cac

DIMENSION

dim_region

- **region_id (PK)**
- region_name
- country
- continent
- market_tier

DIMENSION

dim_channel

- **channel_id (PK)**
- channel_name
- campaign_type
- platform

DIMENSION

dim_campaign_type

- **type_id (PK)**
- type_name
- category
- expected_roi_benchmark

Primary Key (PK) | Foreign Key (FK) | Dimension Table | Fact Table

3. Power Query — Data Cleaning & Transformation

1 Load & Promote Headers

Source CSV → Promoted Headers → Changed Types. Set date as Date, numerics as Decimal/Int.

2 Handle Nulls

Replace null impressions/clicks with 0. Flag null revenue rows with 'Missing Revenue' tag.

3 Remove Duplicates

Table → Remove Duplicates on campaign_id. Keep first occurrence.

4 Standardise Text

Text.Proper() on channel/region. Trim whitespace. Unified casing for CASE sensitivity in DAX.

5 Derive ROI Column

= ([revenue_generated] - [campaign_cost]) / [campaign_cost] → Format as % in model.

6 Derive CTR & CVR

CTR = [clicks] / [impressions] CVR = [conversions] / [clicks] (handle divide-by-zero with if ... then 0)

7 Add Performance Tier

if ROI >= 1 then "High" else if ROI >= 0.2 then "Moderate" else if ROI >= 0 then "Low" else "Loss"

8 Create Date Dimension

List.Dates(start, duration, step) → expand year, quarter, month, weekday columns for time intelligence.

4. DAX Measures — Core Metrics

Every measure with formula and business meaning

Total Revenue

```
SUMX(fact_campaign,
  fact_campaign[revenue_generated])
```

Total attributed revenue across all selected campaigns.

Total Cost

```
SUM(fact_campaign[campaign_cost])
```

Total marketing spend — denominator for ROI and efficiency metrics.

Net Profit

```
[Total Revenue] - [Total Cost]
```

Absolute monetary gain after deducting all campaign costs.

ROI %

```
DIVIDE(
  [Total Revenue] - [Total Cost],
  [Total Cost], 0)
```

For every $1 spent, how many dollars returned as profit. Core KPI.

CTR %

```
DIVIDE(
  SUM(fact_campaign[clicks]),
  SUM(fact_campaign[impressions]), 0)
```

Measures ad creative effectiveness — impression to click rate.

Conversion Rate %

```
DIVIDE(
  SUM(fact_campaign[conversions]),
  SUM(fact_campaign[clicks]), 0)
```

Landing page and offer quality — click to purchase rate.

Avg CAC

```
DIVIDE(
  [Total Cost],
  SUM(fact_campaign[conversions]), 0)
```

Cost to acquire one customer. Lower = more efficient campaign.

MoM Revenue Growth

```
VAR prevMonth =
  CALCULATE([Total Revenue],
  DATEADD(dim_date[date],-1,MONTH))
RETURN DIVIDE([Total Revenue]-prevMonth,
prevMonth, 0)
```

Month-over-month revenue change — drives the trend KPI card arrows.

Revenue Forecast

```
// Enable via Analytics pane
// Line chart → Add Forecast
// Confidence: 95%  Periods: 3
```

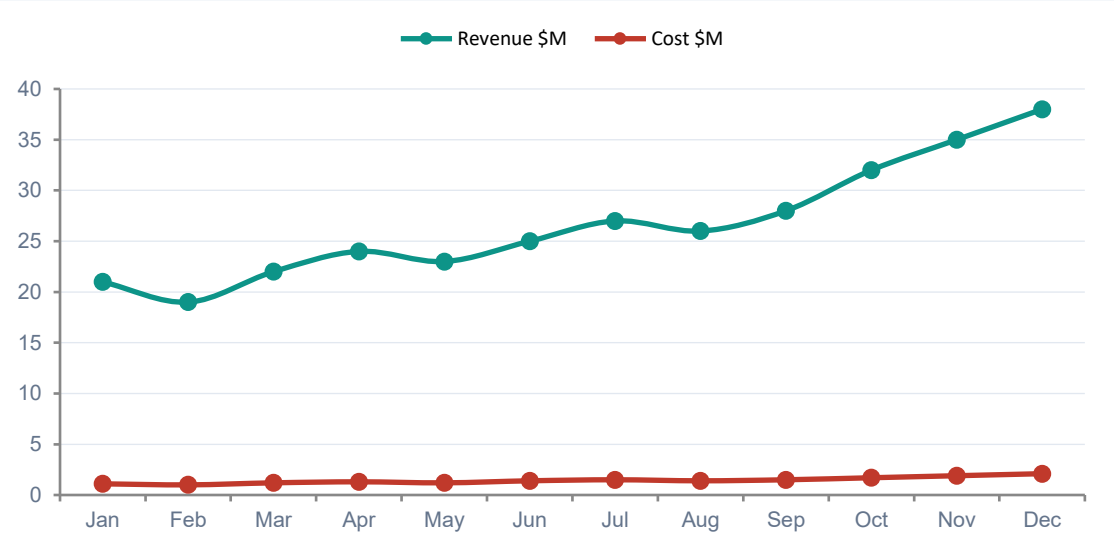
Extends time series 3 months with confidence band — no DAX needed.

5a. Dashboard — Page 1: Executive Overview

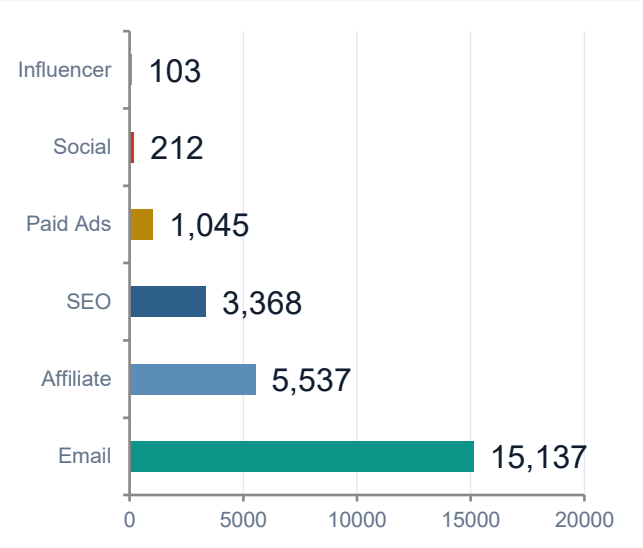
What the CMO and CFO see first — high-level health of all marketing spend

Total Revenue	Total Cost	Net Profit	Overall ROI	Avg CAC
$641M	$32.7M	$608M	1,859%	$283
+12.4% YoY	+8.1% YoY	+14.2% YoY	+4.2pp YoY	−5.3% YoY

Monthly Revenue vs Cost Trend



ROI by Campaign Type

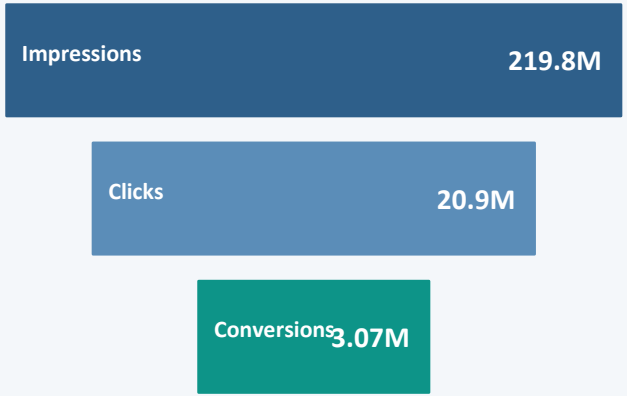


Slicers: Date Range | Campaign Type | Region | Channel

5b. Dashboard — Page 2: Deep Dive Analysis

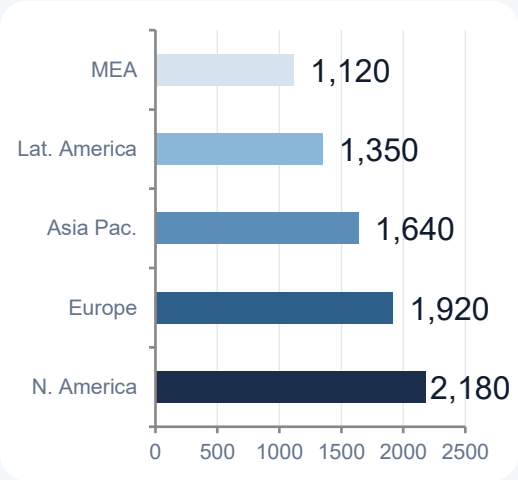
Drill-down by region, channel, and campaign type — for Marketing Manager audience

Conversion Funnel

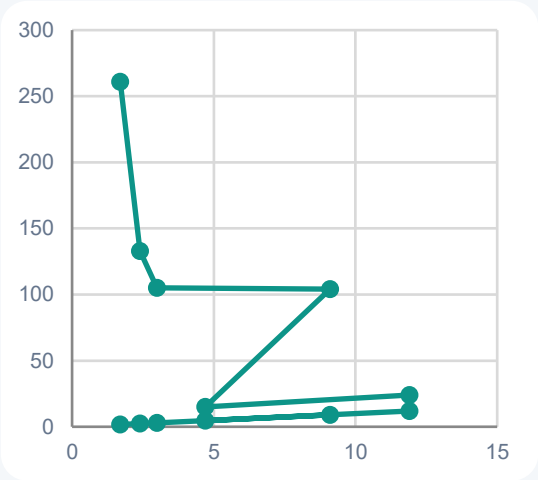


CTR: 9.53% CVR: 6.98% End-to-End: 0.66%

ROI % by Region



Cost vs Revenue (Break-even Line)



Drill-through setup: Right-click any campaign bar → Drill through → Campaign Detail Page. Enables single-campaign deep dive with all KPIs pre-filtered.

Campaign Performance Tiers

High Performer
ROI ≥ 100% → Scale Up

Moderate
ROI 20–99% → Optimise

Low
ROI 0–19% → Monitor

Loss
ROI < 0% → Cut Budget

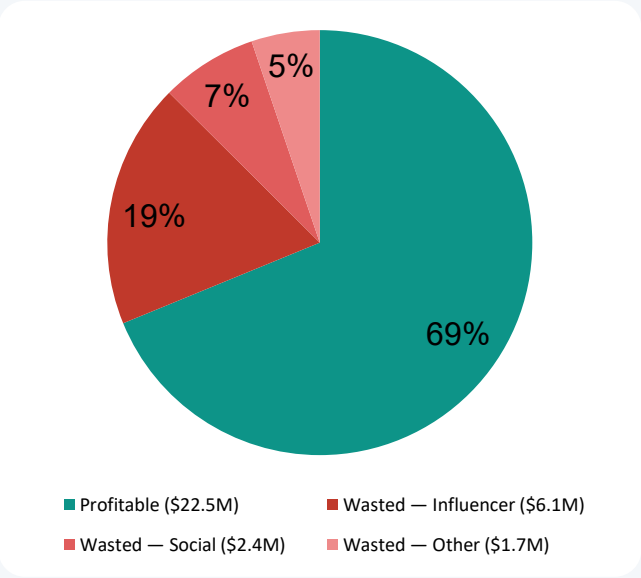
5c. Dashboard — Page 3: Campaign Insights

Top/bottom performers, CAC analysis, and budget waste detection

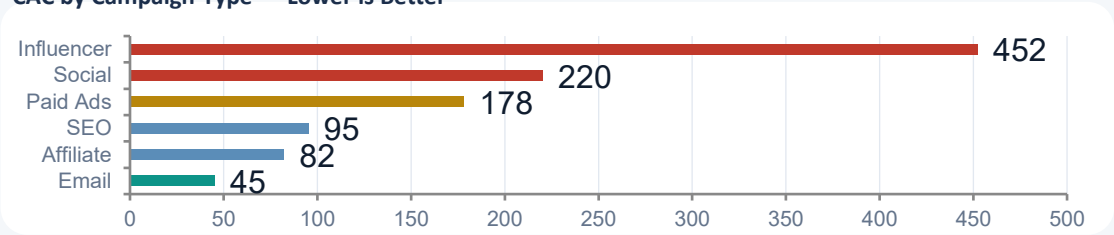
Top 5 vs Bottom 5 Campaigns by ROI

Campaign	Type	ROI %	Action
CAMP-00042	Email	31,200%	**Scale**
CAMP-00187	Affiliate	18,450%	**Scale**
CAMP-00391	SEO	12,300%	**Scale**
CAMP-04832	Email	10,890%	**Scale**
CAMP-03201	Affiliate	9,760%	**Scale**
	— BOTTOM PERFORMERS —		
CAMP-02847	Influencer	−82%	**Cut**
CAMP-04103	Social	−74%	**Cut**
CAMP-01556	Influencer	−68%	**Cut**
CAMP-03899	Social	−61%	**Cut**
CAMP-00782	Influencer	−55%	**Cut**

Budget Waste Breakdown



CAC by Campaign Type — Lower is Better



5d. Dashboard — Page 4: Insights & Recommendations

Actionable findings with projected business impact — decision-ready output

01 Email is the highest-ROI channel

15,137% ROI at $1.7M spend. Gets only 5.2% of budget. Single biggest upside opportunity.

→ Scale +15% budget

02 $10.2M budget completely wasted

31.3% of spend goes to negative-ROI campaigns. Cutting these saves budget with zero revenue loss.

→ Cut 1,250 campaigns

03 Influencer is highest-cost, lowest-CVR

36.3% of budget → 103% ROI. CAC of $450 vs Email's $45. 10× less efficient per customer.

→ Reduce –18% budget

04 Q4 seasonality drives +28% revenue

October–December outperforms Q1 by 28%. Budget allocation does not reflect this pattern.

→ Front-load Q4 +25%

05 Funnel: 90.5% drop at impression→click

CTR improvement of 3pp would add $22M revenue. Creative refresh + tighter audience targeting required.

→ Optimise ad creative

06 Asia Pacific — lowest CAC, highest growth

22% lower acquisition cost than N. America with comparable LTV. Underinvested market with scale potential.

→ Expand AP investment

Projected Total Annual Impact:

$10.2M saved	+$45M revenue	+$22M revenue	+$18M revenue	~$95M total
Cutting waste	Reallocation	Funnel fix	Q4 timing	Combined impact

Project Summary

Marketing Campaign ROI · Power BI Dashboard

Business Problem	Optimise $32.7M marketing spend across 6 channels, 5 regions, 5,000 campaigns
Star Schema	fact_campaign + 4 dimension tables — clean, relationship-driven data model
Power Query	8 transformation steps: null handling, derived columns, date dimension, text standardisation
DAX Measures	9 core measures: Revenue, Cost, ROI, CTR, CVR, CAC, MoM Growth, Forecast, Segmentation
Dashboard Pages	Executive Overview · Deep Dive · Campaign Insights · Recommendations
Business Impact	Identified $10.2M wasted budget; projected $95M annual improvement

Tools: Power BI Desktop · Power Query · DAX · SQL · Python · GitHub